H.T.No. | | | | | | | | | | |

Code No: CT4524 **SRGEC-R23**

III B.Tech I Semester Regular Examinations, November 2025

COMPUTER NETWORKS

(Computer Science and Engineering, CSE(Artificial Intelligence and Machine Learning) & Information Technology)

Time: 3 Hours **Max. Marks: 70**

Note: All questions from **PART-A** are to be answered at one place.
Answer one question from each unit of **PART-B.** All questions carry equal marks.

PART-A

10 × 2 = 20M

1. a) What is the main function of the Physical Layer in the OSI model?
   b) Differentiate between MAN and WAN.
   c) Define error correction. Distinguish between forward error correction and backward error correction.
   d) Differentiate between Go-Back-N ARQ and Selective Repeat ARQ.
   e) What is CDMA? How is it different from FDMA and TDMA?
   f) Mention one feature of 10 Gigabit Ethernet.
   g) How do connectionless and connection-oriented services differ at the network layer?
   h) Name two algorithms for congestion control at the network layer.
   i) Sketch the format of a UDP datagram and describe its fields.
   j) Write any two services offered by UDP.

PART-B

5 × 10 = 50M

UNIT-I

2. a) Consider the following scenario, there are 5 devices, with these devices construct a network with Mesh Topology for the following:
      (i) Draw the Mesh Topology.
      (ii) For each device, how many dedicated lines requested to connect with other devices?
      (iii) Totally, in Mesh Topology how many duplex - mode links needed? (5M)
   b) Explain the structure and working of fiber optic cables. (5M)

(OR)

3. According to ISO standard what are the different Layers in the OSI Reference Model? llustrate the Functionalities of each layer with the help of the layered architecture. (10M)

UNIT-II

4. Analyze, how the Data Link Layer ensures reliable communication using elementary protocols. (10M)

(OR)

5. a) Describe Cyclic Redundancy Check (CRC) with a neat diagram. (5M)

   b) Explain the checksum method for error detection. (5M)

UNIT-III

6. a) Explain Ethernet protocol in detail, including frame format, addressing, and error detection. (5M)

   b) Compare the throughput of pure ALOHA and slotted ALOHA. (5M)

(OR)

7. a) What is CSMA? Bring out the differences between 1-persistent, non- persistent, and p-persistent CSMA? (5M)

   b) With diagrams, illustrate how multiple users share a single frequency channel in TDMA and CDMA. (5M)

UNIT-IV

8. a) Analyze the main routing algorithms used in networks and compare their functions. (6M)

   b) Provide an example of the optimality principle in shortest path calculation. (4M)

(OR)

9. a) Why is subnetting necessary for modern networks, and how does CIDR address this need? (6M)

   b) Discuss the contrast between classful and classless addressing schemes. (4M)

UNIT-V

10. a) Draw and label the components of a UDP datagram; explain their purposes. (5M)

    b) Select and discuss two UDP-enabled applications. (5M)

(OR)

11. a) Describe the working of HTTP and its key features. (6M)

    b) Explain the differences between HTTP 1.0 and HTTP 2.0. (4M)

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